College of Computing

Department of Data Science

Deep Learning-Based Modeling and Forecasting of Greenhouse Gas Emissions

Group Name: GROUP 4

Group Members

No	Name	ID No
1	Dagmawi Legesse	DBU1501102
2	Melis Melakie	DBU1501357
3	Haymanot Tadese	DBU1501258
4	Yosef Getaneh	DBU1501589
5	Esayas Melaku	DBU1501174
6	Kenean Bisrat	DBU1501301
7	Wudasiemariam Muluken	DBU1501537

Submission To: Baye Y, PhD

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Contents

1 Introduction

Urbanization is a process in which the development of a particular region takes place. This process involves rapid population growth, an increase in the proportion of the non-agricultural workforce, and changes in land use from agricultural to non-agricultural patterns [1]. In this process of urbanization, many industries are introduced, which in turn help create large-scale employment, much needed in developing countries. However, the industrial revolution has played a significant role in emitting greenhouse gases over the past two centuries because human activities led to the use of machines and the mechanization of processes that were previously performed by hand. Technological innovations, rapid economic growth, territorial expansion, rapid population growth, the emergence of urban areas, and the transformation of the global scientific system are leading to the beginning of the industrial revolution [2].Because of that, industrial practice emits greenhouse gases that lead to climate change and global warming.

Ethiopia has experienced rapid urbanization, increased industrialization, and expanding economic activities, all contributing to a significant rise in fuel consumption and greenhouse gas (GHG) emissions, such as carbon dioxide (CO2), methane (CH4), nitrous oxide (N2O), and others, from the industry sector. The increase in GHG concentrations as a result of human activity enhances the greenhouse effect, leading to global warming. This process influences the climate, causing changes in weather patterns, sea levels, and other aspects of the environment [3]. CO2 is the largest contributor to GHG emissions; therefore, an essential part of environmental control and climate change policy measurement. Such concrete predictions allow governments, certain industries, and environmental agencies to have predictions on what may happen in the coming years, formulate their future policies on how to approach the issue of climate change, and decide on how to reduce the impact of climate change in the world [4].

There are different related studies conducted on GHG emissions using statistical methods and Machine learning techniques. The high CO2 emissions raise the need to employ modern techniques in predicting CO2 emissions since the traditional statistical methods fail to capture the nature and the source of variation in emission levels and the availability of high continuous, high dimensionality, complex, uncertain and unstructured, and dynamic timeseries data, machine learning models typically rely on manual feature engineering, requiring domain expertise to extract meaningful representations from raw data. This process is often time-consuming, task-specific, and may fail to capture complex non-linear relationships present in large-scale datasets [5]. this study, conducted with Deep learning models, is one of the most promising computational techniques with multiple hidden layers, addressing the machine learning limitations by enabling automatic feature learning and extraction, and is best for forecasting time series data, making the prediction process faster and more efficient than machine learning techniques. As [6, 7, 8]. it is noted that deep learning models for time series data are preferable when it comes to forecasting CO2 emissions because they can reflect both short-term and long-term dynamics of the emissions' fluctuations.

This research conducts LSTM, GRU, and LSTM-GRU deep learning algorithms to study the forecasting of GHG emissions, with an analysis of their performance based on the best predicted model, which helps policymakers make better decisions and implement effective environmental strategies to mitigate climate change problems in industries [9].

2 Statement of the Problem

The main problems we tried to see and solve in this research are the lack of a predictive and functional model for predicting GHG emissions at large factories and industries in Ethiopia. While different data sources exist that have GHG emission data, like the CEIC data, which have country-level emissions, there is no equivalent to the U.S. GHGRP facility-level emission data reporting that captures all industrial and factory GHG emission variables. This data problem limits the ability to model and predict emissions accurately and identify high-emitting factories that affect the creation and implementation of data-driven emission monitoring strategies and policies. As Ethiopia is undergoing rapid industrialization, the government is shifting its focus to an industry-led economy rather than an agricultural-led one; therefore, the need for a reliable and robust prediction model is needed to support climate-aware development and meet international responsibilities Ethiopia acquired when signing different environmental treaties. By training deep learning models in the U.S. GHG emission data that has variables like unit type, Heat capacity, and other emission-determining variables, this project creates a model that can be transferred and implemented in Ethiopia. The developed prototype allows users to input local data (e.g., facility-level records in Excel/CSV) to generate predictions.

Nowadays, the advancement of technologies, such as Artificial Intelligence, the Internet of Things (IoT), Machine Learning, Deep Learning, Data Analysis tools, and 5G Wireless Networks, is the fundamental enabler of this concept. The concentration of human populations into discrete areas leads to the transformation of land for residential, commercial, industrial, and transportation purposes. Due to this, poor air and water quality, insufficient water availability, waste-disposal problems, and high-energy consumption are exacerbated by the increasing population density and demands of urban environments. The most pressing problems facing the world today also come together: poverty and environmental degradation. Different research conducted on Machine Learning, combined with IoT data, played an important role in different domains of smart cities, e.g., mobility, environment, security, and healthcare. But now with the advancements of AI, ML faces challenges on data quality and imbalance, computational demands, overfitting, and bias in the output that leads to having higher complexity for understanding results for decisions.

Therefore, designing effective models to accurately predict greenhouse gas emissions using deep learning is very crucial to control those machine learning model problems. Ethiopia is one of the emerging industrial hubs of Africa; it is a preferred destination for local and international people and investors attracted by the peace, unlimited natural resources for industrial inputs, and diverse and popular customer handling capabilities. Nowadays, there are tens of thousands of different manufacturing industries in Ethiopia. This rapid urbanization and industrialization have led the city into a new era of climate change and are seen as a modern-day curse. According to [10] the increasing industrial sector capacity leads to carbon emissions, necessitating a low-carbon transformation for global sustainability. Accurate time-series forecasting of industrial carbon emissions is crucial for optimizing production, reducing energy consumption, improving efficiency, and achieving decarbonization. However, current forecasting faces challenges like model uncertainty and low accuracy with multiple forecasting steps. To address these issues, this study proposed a hybrid deep learning carbon emissions forecasting model for industrial processes.

3 Objective

Before explaining the objective, we must see the problem caused by greenhouse gas emission greenhouse gas emission has many problems in climate change, drives global warming and affects many areas such as health, agriculture and economy, so government and policymakers want to know about future emission to prepare for the change and make environmental policy, but unfortunately countries does not have enough data or tools to make predictions and prepare for future change, so decisions are made using traditional methods and incomplete information, so our project aims to solve this problem .

The main goal of this project is to develop a deep learning model that predicts greenhouse gas emission using historical data from the United States, the project focuses on learning patterns and predict future emission and also our prototype accepts csv file and then generates future emission predictions.

3.1 Specific Objectives

- To analyze industrial greenhouse gas emissions using deep learning techniques.
- To develop and compare LSTM, GRU, and hybrid deep learning models.
- To identify the best-performing model for time-series GHG forecasting.
- To design a functional prototype for real-world emission prediction.

4 Project Scope

Our project focuses on predicting future emission of greenhouse gases from industries using deep learning models. Our model was trained on the U.S Green house gas reporting program data and make it apllicable in Ethiopia.

5 Variable Explanation

The key variables used are explained below.

5.1 Facility and Location Variables

- **Facility ID**: A unique numerical identifier assigned to each facility. It is used as unique identfier.
- **Facility Name**: The registered name of the facility reporting emissions.
- **City**: The city in which the facility is located.
- **State**: The U.S. state where the facility is found.

5.2 Industry Classification Variables

- **Primary NAICS Code**: The North American Industry Classification System (NAICS) code representing ZIP codes, phone area codes, or product category IDs.
- **Industry Type (Sectors)**: An overview classification of the working sector (e.g., manufacturing).
- **Industry Type (Subparts)**: A more detailed classification that specifies industrial process type or emission subcategory within a sector.

5.3 Temporal Variable

- **Reporting Year**: The calendar year in which the emissions data were reported.

5.4 Unit and Capacity Variables

- **Unit Name**: The variables tells emissions-generating unit in a facility (e.g., boiler, turbine).
- **Unit Type**: The classification of the unit based on its function.

- **Unit Reporting Method**: The reporting method of emission by facilities that is put in this dataset for regulatory compliance.
- **Unit Maximum Rated Heat Input Capacity (mmBTU/hr)**: The maximum heat input capacity of the unit measured in million British Thermal Units per hour. This variable represents the operational scale of the unit and is a key predictor of emission magnitude.

5.5 Emission Variables

- **Unit CO_2 Emissions (Non-Biogenic)**: Carbon dioxide emissions excluding biogenic sources.
- **Unit Biogenic CO_2 Emissions**: Carbon dioxide emissions from biological sources such as biomass combustion. These emissions are often reported as zero or unknown.
- **Unit Methane (CH_4) Emissions**: Methane emissions produced by industrial activities. Methane has a significantly higher global warming potential than CO_2.
- **Unit Nitrous Oxide (N_2O) Emissions**: Nitrous oxide emissions generated during industrial processes. Although emitted in smaller quantities, N_2O has a very high global warming potential.

6 Project Methodology

6.1 Preprocessing

First step was loading our data set, as it was in excel (`xlsb`) format we used the `pyxlsb` to load it..

The first step of preprocessing was checking for duplicated and null values. We found no duplicated values but we found null values on different variables so we had to handle them. In the handling step we had dropped irrelevant variables due to redundancy and lack of predictive power, we also carried out imputation the imputed values and with what the null was replaced is as follows:

- **Primary NAICS Code**: Imputed as *Unknown*
- **Unit Biogenic CO_2 Emissions**: Imputed as 0 because biogenic CO2 is often zero or unreported
- **Unit Maximum Rated Heat Input Capacity**: We imputed it by group median for values that had unit type and for null unit types global median was used

- **Unit Type**: Imputed as *Unknown*

The final preprocessing step was feature engineering the first engineered variable was CO_2 equivalent (CO_2e) equivalent it was created because the effect of the three green house gases was not the same because of that the model might give a bigger weight to a green house gas not based on effect but based on values so to solve that we had to created a variable that was equivalent to one green house gas that can be used as a target and that can capture the full effect of the three to do that carbon di oxide was selected and the calculation was carried out as follows:

$$\text{Total CO}_2\text{e} = \text{CO}_2 + \text{Biogenic CO}_2 + 25 \times \text{CH}_4 + 298 \times \text{N}_2\text{O}$$

Then another variable was created called `Series_ID` we created a series ID because there are many facilities and Each facility has many units also Each unit has values for different years so to track those Facility Id and Unit Name are not enough so we created a new variable that combines the two to fully identifies one physical emissions source. `Is_Zero_Reported` was engineered to flag (binary for zero scaled CO_2e) to make the model learn non-emission years and make more accurate predictions

6.2 Feature Transformation, Encoding, and Scaling

The data must be check for outlier and the distribution also had to be check. When checking we discovered that we had outliers and a very high right skewness this problem had to be solved. To fit it we used a transformation called Yeo-Johnson transformation and we applied it to our numeric variables Total CO_2e and Unit Maximum Rated Heat Input Capacity (mmBTU/hr).

Yeo–Johnson Transformation

The Yeo-Johnson Transform is a statistical method used to stabilize variance and make data more closely resemble a normal distribution. It is an extension of the Box-Cox Transform, but unlike Box-Cox, it can handle both positive and negative values. This makes it especially useful in data preprocessing for machine learning models that require normally distributed or symmetrically distributed data, such as linear regression or principal component analysis (PCA). [10]

The Yeo–Johnson transformation is defined as:

$$T(y;\lambda) = \begin{cases} \dfrac{(y+1)^{\lambda}-1}{\lambda}, & y \geq 0,\ \lambda \neq 0 \\ \ln(y+1), & y \geq 0,\ \lambda = 0 \\ -\dfrac{(-y+1)^{2-\lambda}-1}{2-\lambda}, & y < 0,\ \lambda \neq 2 \\ -\ln(-y+1), & y < 0,\ \lambda = 2 \end{cases}$$

Then scaling was carried out with MinMaxScaler (0-1 range). Scaling is used because we have large emission values and the extremely large values will dominate the small values as the model understands numbers. For the domination not to happen, we used MinMaxScaler, which scales them in the range of 0 to 1. We chose this scaling rather than the standard scaler because standard scaler scales the data in the range of -1 to 1, we used the ReLU activation function if it receives a negative input, the neuron will become dead and information loss will happen to prevent that MinMaxScaler, which scales the data in the range between 0 and 1 is used. Then encoding categorical columns (State, Industry Type (sectors), Unit Type, Primary NAICS Code) was carried out using embedding ID mapping. This encoding wont add extra dimension because of that it is suitable.

6.3 Sequence Generation

The most important part in our model input preparation step was the sequence generation part. Here we first separated the features as follows:

1. **Dynamic feature** – These features are feature which change based on time. They are Total_CO_2e_Scaled, Capacity_Scaled, and Is_Zero_Reported.
2. **Static feature** – These are feature that doesn't change over time. They are the categorical columns.
3. **Target** – The feature we out trying to predict for the next year it is Total_CO_2e_Scaled.

Then sequence was generated using the sliding window technique. The sliding window approach in time series forecasting is a technique that structures sequential data into fixed-length input-output pairs to train models. Instead of treating the entire time series as a single sequence, the method creates smaller, overlapping windows of past observations (inputs) and corresponding future values (targets) [11].

For our sequence a window (lookback) of 6 was chosen because it was the optimal value that had the best autocorrelation between years, and that generated enough sequence for our model.

Finally, data was split into a 70% train, 10% validation, and a 20% test set and it was given to the model.

6.4 Architecture

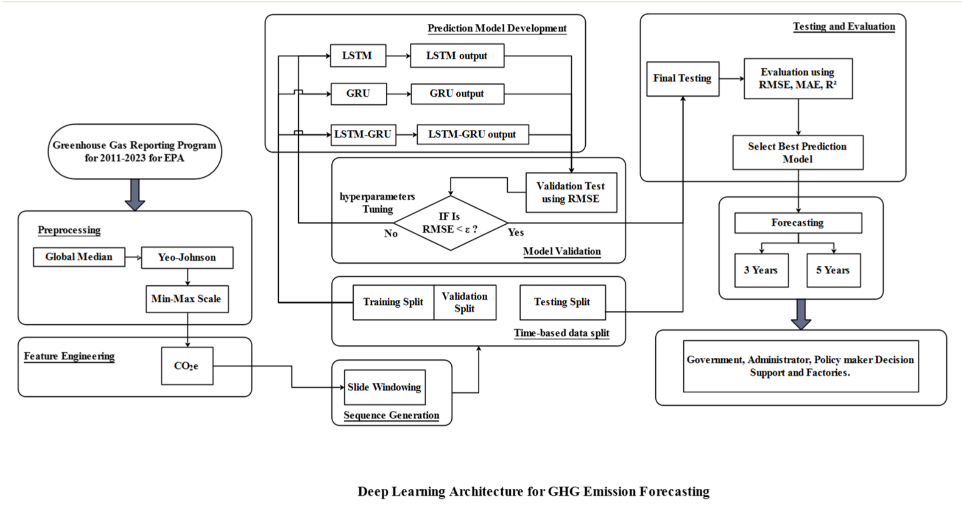


Figure 1: Flow chart

The flow chart is presented now let's discuss about the models

Recurrent neural networks (RNNs) are designed for processing sequential data, such as text, speech, and time series where the order of elements is important. Unlike feedforward neural networks which process inputs independently, RNNs utilize recurrent connections, where the output of a neuron at one time step is fed back as input to the network at the next time step. This enables RNNs to capture temporal dependencies and patterns within sequences. [12]

While it is effective for shorter sequences, RNNs struggle with long-term dependencies due to the vanishing gradient problem, where gradients diminish as they are backpropagated through many time steps, causing the model to forget earlier information in long sequences. To address this, Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU) were developed. Both LSTM and GRU are special kinds of Recurrent Neural Network (RNN) with distinct architectural differences. Both of them capture long term dependencies in sequential data by utilizing specialized gating mechanisms that control the flow of information. The intuition is to create an additional module in a neural network that learns when to remember and when to forget pertinent information. [13]

For this project we used a hybrid deep learning LSTM and GRU models then another Hybrid model was created. We developed the models using TensorFlow/Keras. Finally, they were evaluated by different metrics and the best performing model was selected for predictions. The model is as follows:

1. **LSTM:**

   Dynamic branch (Dynamic feature):

- LSTM layer with 64 units
- LSTM layer with 32 units

Static branch (Static feature):

- Dense layer with 16 neuron, ReLU activation

Fusion and output:

- Concatenate layer with 48 neurons to combine dynamic and static branch outputs
- Dense layer with 16 neurons, ReLU activation to extract the fusion features
- Final dense layer with 1 Neuron for output

Training:

- Optimizer: Adam (lr = 0.0001)
- Loss: Huber
- Metric: MAE

2. **GRU**

   Dynamic branch (Dynamic feature):

   - GRU layer with 64 units
   - GRU layer with 32 units

   Static branch (Static feature):

   - Dense layer with 16 neurons, ReLU activation

   Fusion and output:

   - Concatenate layer with 48 neurons to combine dynamic and static branch outputs
   - Dense layer with 16 neurons, ReLU activation to extract the fusion features
   - Dropout layer (rate = 0.1) randomly set 10% neuron to 0 to prevent overfitting
   - Final dense output layer with 1 Neuron for output

   Training:

   - Optimizer: Adam (learning rate = 0.0001)
   - Loss: Huber

- Metric: MAE

3. **Hybrid**

   Dynamic input branch (Dynamic feature):

   - **LSTM sub-branch:**
     - LSTM layer with 64 units
     - LSTM layer with 32 units
   - **GRU sub-branch:**
     - GRU layer with 64 units
     - GRU layer with 32 units

   Static input branch (Static feature):

   - Dense layer with 16 Neuron, ReLU activation

   Feature fusion:

   - Concatenate outputs from LSTM, GRU, and static branches
   - Dense layer with 32 Neuron, ReLU activation
   - Dropout layer with rate 0.2 that randomly set 20% neuron to 0 to prevent overfitting
   - Dense layer with 16 Neuron, ReLU activation

   Output layer:

   - Dense layer with 1 Neuron for output

   Training:

   - Optimizer: Adam (learning rate = 0.0001)
   - Loss: Huber
   - Metric: MAE

GRU performed best (highest $R^2 \sim 0.93$–0.95, lowest MAE/RMSE on validation). but the change in performance between the LSTM and GRU was very small, so we had to check for the significance and found no significant difference from hybrid via the Diebold-Mariano test).

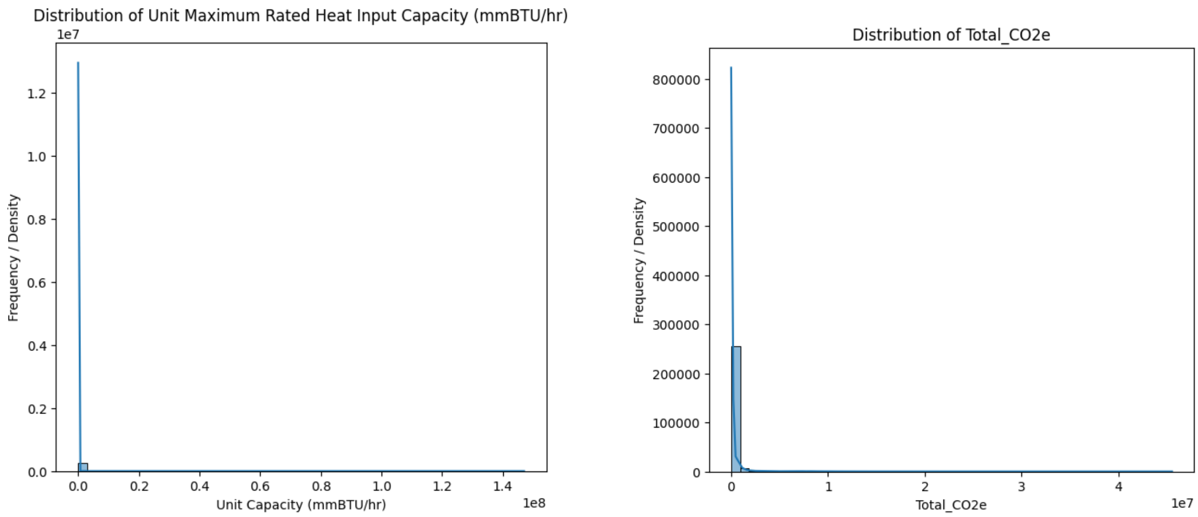
6.5 Forecasting and Prototype

Forecasting was carried out using the GRU model which was our best model out of the 3 that we tried. So as GRU is and autoregressive model this is a type of time series model where future values are predicted based on a linear combination of past values of the same variable. The idea is simple: "The future depends on the past." [14]. We did a five-year forecasting by iteratively predicting the next year using it again as an input by the sliding window mechanism and predicting the other year. The new outputs are numbers that are transformed using the Yeo–Johnson transformation and scaled using the min-max because of that, the output is not the real values rather scaled and transformed small numbers. To fix that we applied inverse transformation to obtain real-world CO_2-equivalent emissions in metric tons.

Finally, to make the project real-world applicable, we made a functional prototype that allows users to input new emission data via a CSV file. Then the prototype loads all the artifacts (model, scaler, transformation parameters, sequence and feature configurations), applies the same preprocessing steps automatically and generates a next-year emission forecast. Our prototype handles missing years by using the backfill (or padding) filling method as our sequence requires a 6-year emission data for prediction this method fill the missing years with the smallest years data.

7 Discussion and Results

In our exploratory analysis confirmed high skewness in U.S. emissions data and a large number of outliers



(a) Unit heat skewness

(b) CO_2e skewness

Figure 2: Distribution skewness of unit heat input capacity and total CO_2e emissions

Total_CO_2e

- Q1: 215.60
- Q3: 63190.67
- IQR: 62975.07
- Lower Bound: -94247.00
- Upper Bound: 157653.27
- Min Value: 0.00
- Max Value: 45538010.37
- Outliers: 41837

Unit Maximum Rated Heat Input Capacity (mmBTU/hr)

- Q1: 8.00
- Q3: 91.40
- IQR: 83.40
- Lower Bound: -117.10
- Upper Bound: 216.50
- Min Value: 0.00
- Max Value: 147000000.00
- Outliers: 33827

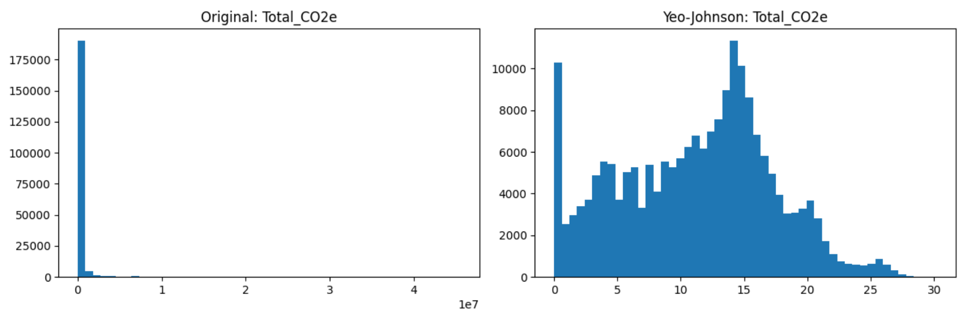


Figure 3: After transformation

Also, in our EDA we found that emission was decreasing in USA

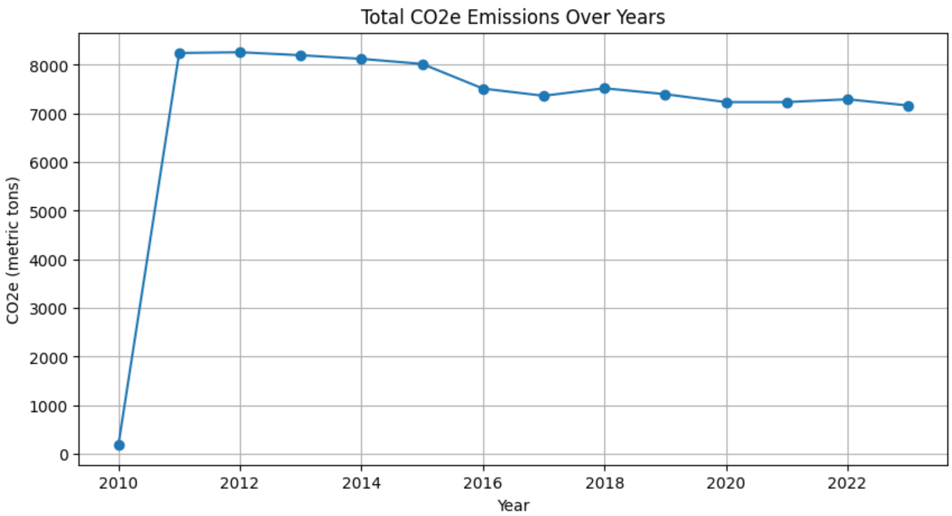


Figure 4: Emssion over the years

And we confirmed that indeed it was decreasing as our prediction show that from 2023 in the next 5 years emission will decrease by 1,758,686,822 Metric tons.

When comparing the three model the Hybrid model was the lowest performer when evaluationg using the MAE, RMSE and R^2

Table 1: Performance comparison of deep learning models

Model	MAE (Scaled)	RMSE (Scaled)	R^2	Inference Time (s)	Parameters
GRU	0.0265	0.0506	0.9369	10.70	23,537
LSTM	0.0280	0.0524	0.9322	9.71	30,705
Hybrid	0.0525	0.0670	0.8892	10.48	55,697

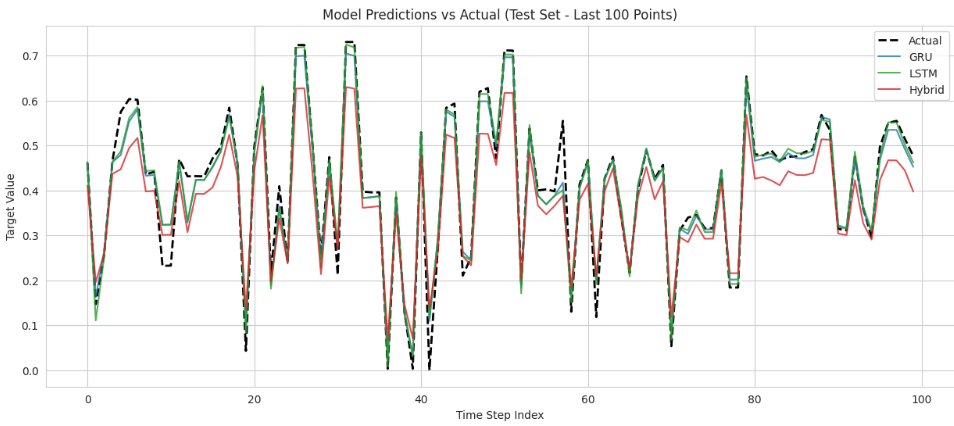


Figure 5: Models performance

The GRU and LSTM models achieved better performance: $R^2 \approx 0.94$ and $R^2 \approx 0.93$, low MAE/RMSE on scaled data. We also see visually that both the LSTM and GRU models fit the data well compared to the Hybrid model which has a very bad performance. The Hybrid model consistently fails to accurately predicts peak values mostly it under predicts them, whereas the LSTM and GRU models demonstrate high accuracy in tracking and predicting rapid fluctuations and extreme values within the test set.

So, to see if the low performance changes between the LSTM and GRU model is a coincidental change or if it is statistically significant we used (Diebold–Mariano) test this test starts with the null hypothesis stating the two models have equal predictive accuracy then it calculates the p-value and tells if it is significant or not. The interpretation shows that we had a p-value < 0.05 proving in fact the GRU model's performance being better is statistically significant.

8 Conclusion and Future Work

This project successfully developed a robust deep learning-based GHG emissions forecasting model, trained on the U.S. greenhouse gas emission data, that helps in aiding policymakers and addressing predictive model gaps in Ethiopia. Future works include refining the model and applying it on Ethiopian facility emission data by collecting different emission data from different sources (e.g., from national inventories), incorporating additional variables that influence factories' greenhouse gas emission (e.g., GDP), and creating a website for the prototype to be accessible by different facilities.

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